CAPSTONE DEFENSE BRIEF | NOVEMBER 2026

OCR Benchmark

What the test checked, what it found, and what the results do - and do not - tell us.

The result in one sentence

In a controlled test of 100 generated school-document PDFs, Apple Vision read names and printed student numbers with useful accuracy, and the OCR-to-matching flow completed quickly; rotated pages were the main weakness.

100 / 100	80 / 80	84 / 100	159 ms
documents processed successfully	printed student numbers read exactly	correct student candidates found	average end-to-end time

What we tested

We ran the system OCR on 100 generated PDF samples: 20 each of transcripts, registration certificates, Form 137 / SF10 records, student IDs, and PSA birth certificates. Each document type was tested across five quality levels, from clean files to faded or altered text.

Measure	Observed result	Plain-language meaning
Name text errors	CER 0.160; WER 0.160	On average, about 16 in every 100 reference characters or words differed in the extracted name text.
Printed student-number reading	80 of 80 exact	All 80 documents that displayed a student number were transcribed exactly. Birth certificates without printed numbers were excluded.
Finding the intended student	84% recall; 100% precision	The intended student appeared in 84 of 100 candidate lists. No unrelated student was suggested in 1,900 distractor checks.
Processing time	158.6 ms average	The complete OCR request, name processing, and matching flow was below the 3,000 ms target.

How to explain the two name scores

CER and WER measure text transcription. The 84% recall score measures a separate step: whether the system found the intended student from the extracted name. A correct number can still be read even when a name match is missed, so these scores answer different questions.

RESULTS IN CONTEXT

What worked - and where it struggled

Strengths

The software completed all 100 OCR requests without a failure. It read every printed student number exactly in the 80 documents that had one. In the roster-candidate check, it did not suggest an unrelated student across the 1,900 distractor opportunities in this test.

Main weakness: rotated pages

The most notable drop happened in Q3, where pages were rotated a full 90 degrees. In that group, name transcription error rates rose to 0.800, and only 4 of 20 intended students were found. Printed student numbers still read exactly in all 16 Q3 samples that displayed a number. This points to a weakness in reading and matching names on severely rotated pages.

Document condition	Name text error rate (CER / WER)	Correct student found
Clean / small text / speckled / faded (Q1, Q2, Q4, Q5)	0.000 / 0.000	20/20 for Q1, Q2, Q4, and Q5
90-degree rotation (Q3)	0.800	4/20

Presenter note

Describe the 90-degree result as a stress-test finding. Q3 was a deliberately rotated generated page; it does not represent ordinary scanner feeder skew or prove the same error rate on real student submissions.

How to say the timing result

The measured average was 158.6 milliseconds from the authenticated OCR request through name processing and student matching. That is below the project benchmark target of 3,000 milliseconds. This is a measured result on the tested setup, not a guarantee for every computer or network.

Suggested conclusion

> **The benchmark shows that the current OCR flow can process these generated samples quickly and read printed student numbers reliably. Name-based student matching was less reliable, especially when pages were rotated, so the system should keep staff review in the workflow when OCR results are uncertain.**

DEFENSE SPEAKING GUIDE

A simple 45-second explanation

“We tested the PUPSJ-RMS document OCR using 100 generated PDF samples covering five school document types and five document-quality conditions. On the Mac test setup, all 100 samples were processed. The system read all 80 printed student numbers exactly. It found the intended student in 84 out of 100 name-matching cases, with no unrelated candidate returned in 1,900 checks. The full process averaged about 159 milliseconds, below our three-second target. The main weakness appeared on pages rotated 90 degrees, where name matching dropped. Since these were generated test documents rather than physical scans, the results are an initial benchmark and should be followed by testing with real, approved sample scans.”

Acronym and term legend

Term	Meaning you can say aloud
OCR	Optical Character Recognition: software that reads text from a scanned or image-based document.
CER	Character Error Rate: how often individual characters differ from the expected text. Lower is better.
WER	Word Error Rate: how often words differ from the expected text. Lower is better.
Recall	Of all intended students, the share that appeared in the system's suggested candidate list.
Precision	Of all suggested student candidates, the share that was correct.
Candidate / roster matching	The step that compares an extracted name with student names and returns possible matches.
Apple Vision	Apple's built-in text-recognition framework; this benchmark ran it on macOS.
PDF	Portable Document Format, the file format used for the test documents.
SF10	School Form 10, the learner's permanent academic record, formerly called Form 137.
PSA	Philippine Statistics Authority.
Q1-Q5	Five generated document conditions: clean, smaller text, 90-degree rotation, speckle/streak marks, and faded print.

Scope to state if asked: This was a controlled benchmark using generated documents, not a field study using physical scans. Native results cover macOS Apple Vision only. Windows and Linux OCR performance were not tested. The quality variations were synthetic; Q3 was a 90-degree rotation. Student-number exact match used only the 80 samples where a number was printed.